11.4 The Tautochrone

WORKSHEET

The question

STEP III 1994, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/94-s3-q9

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself The bowl is smooth. Which forces act on the particle, and which of them does work?

STAGE 2 Simple harmonic motion in s

Ask yourself Use $s = 2\sqrt{ky}$ to replace y by a function of s. What kind of motion does the resulting equation describe?

STAGE 3 The time to the bottom

Pause and try Why does this work? The "restoring force" along the curve is g times the slope, $g\frac{dy}{ds} = \frac{g}{2k}s$: exactly proportional to the distance from the bottom, measured along the curve. That is the defining property of simple harmonic motion. A particle starting higher has further to go, but is pushed proportionally harder.

STAGE 4 **Two particles**

Ask yourself When do the particles reach V, and how fast are they moving? Who has more momentum?

Ask yourself With total momentum zero and coefficient of restitution α, what are the velocities just after the collision?

What's really going on

Why the cycloid?

A wheel of radius R rolling under a line traces the cycloid $x = R(t + \sin t)$, $y = R(1 - \cos t)$. Then $\frac{\mathrm{d}x}{\mathrm{d}t} = R(1 + \cos t)$ and $\frac{\mathrm{d}y}{\mathrm{d}t} = R\sin t$, so the speed along the curve is $R\sqrt{2 + 2\cos t} = 2R\cos\frac{t}{2}$. Integrating, the arc length from the bottom is $s = 4R\sin\frac{t}{2}$, while $y = 2R\sin^2\frac{t}{2}$. Hence $s^2 = 8Ry$: the cycloid is exactly the curve with $s = 2\sqrt{ky}$, with $k = 2R$. Conversely, the requirement that the restoring force be proportional to s forces the curve to be a cycloid.

The brachistochrone

In 1696 Johann Bernoulli challenged the mathematicians of Europe: along what curve does a bead slide from one point to a lower point (not directly below it) in the *shortest* time? The answer, the *brachistochrone*, is again a cycloid. Newton is said to have solved the problem in a single evening after receiving it; he published his solution anonymously, but Bernoulli recognised the author, "as the lion by its claw". The problem is the founding example of the *calculus of variations*, which you met in Section 4.7: finding the curve that minimises a quantity defined by an integral.

Huygens's cycloidal clock

Huygens knew that a simple pendulum keeps imperfect time because its period depends on its amplitude. In his *Horologium Oscillatorium* (1673) he showed how to make the bob move along a cycloid: hang the string between two curved "cheeks" shaped as cycloids, and the string wraps round them in such a way that the bob traces a cycloid itself (the *involute* of a cycloid is another cycloid). In practice the cheeks caused more error through friction than they removed, and clockmakers instead kept the swings small. But Huygens's analysis invented the theory of evolutes and involutes, and his book was one of the great works of seventeenth-century mechanics.

Try this next

1. **Different heights.** A particle is released from height h in the tautochrone bowl. Show that its speed at V is $\sqrt{2gh}$, as it must be by energy conservation, and check that this agrees with ωs_0.

2. **An ordinary bowl.** In a hemispherical bowl of radius R, compare the time taken to reach the bottom from a small height with the time from the rim. Which is longer? (Use the result of Section 11.3.)

3. **Equal masses.** Repeat the second part of the question with two particles of equal mass m, released from heights h_1 and h_2, and coefficient of restitution e. Do they still always collide at V?

4. **Cycloid arc length.** Show that one arch of a cycloid has length $8R$, and that the area under it is $3\pi R^2$, three times the area of the rolling circle.

Further reading

- **George Simmons, *Differential Equations with Applications and Historical Notes*** (CRC Press). Includes the tautochrone and the brachistochrone, with their history.
- **E. A. Whitman, "Some historical notes on the cycloid"**, *American Mathematical Monthly* 50 (1943). Why the cycloid was called "the Helen of geometers".
- **Paul Nahin, *When Least Is Best*** (Princeton University Press). The brachistochrone and other optimisation problems, told with history.

The full guided solution is in Section 11.4:
https://beyondstep.keithmcnulty.org/ch11/tautochrone.html